

Yunus Serhat Bıçakçı, PhD

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Istanbul, Türkiye;; Glasgow, UK

Research profile

Researcher and AI engineer with a PhD in Applied Informatics, specialising in geospatial artificial intelligence, multimodal learning, vision-language models, large language models, retrieval-augmented generation, computer vision, semantic segmentation, object detection, and spatial data science. My recent work focuses on scalable AI workflows that connect street-level imagery, remote sensing, language models, and geographic information systems for urban analytics, environmental criminology, and GeoAI applications.

Research interests

Geospatial Artificial Intelligence, Vision-Language Models, Large Language Models, Retrieval-Augmented Generation, Multimodal Learning, Computer Vision, Remote Sensing, Semantic Segmentation, Object Detection, Spatial Data Science, Geographic Information Systems, Urban Analytics, Environmental Criminology, Natural Language Processing

Positions

2026 – Present

Assistant Professor

Marmara University, Department of Artificial Intelligence and Machine Learning, Faculty of Applied Sciences – Istanbul, Türkiye

Conduct research and teaching in artificial intelligence, machine learning, computer vision, geospatial AI, multimodal learning, and spatial data science. Develop research outputs on vision-language models, large language models, retrieval-augmented generation, semantic segmentation, object detection, and urban-scale geospatial analysis. Contribute to departmental development, curriculum design, research collaboration, and externally funded project activities.

2024 – Present

Research Software Engineering Scientist / Affiliate

University of Glasgow, Geospatial Data Science Group, School of Geographical and Earth Sciences – Glasgow, UK

Initially appointed as a Research Associate on the Multi-LM project and subsequently continuing as an affiliate research software engineering scientist within the Geospatial Data Science Group. Contribute to collaborative projects on geospatial data science, multimodal AI, street-level imagery, LLM embeddings, geospatial foundation models, and urban analytics. Co-author scholarly publications, develop research software pipelines, and present findings at international conferences.

2023 – 2024

Postdoctoral Researcher

University College London (UCL), Jill Dando Institute of Security and Crime Science – London, UK

Conducted postdoctoral research on AI-driven crime mapping, geospatial analysis, natural language processing, and large-scale urban data analytics. Developed machine learning workflows for crime and urban data analysis, produced scholarly outputs, and presented findings at international conferences and seminars.

2022 – 2023

Assistant Professor

Marmara University, Vocational School of Social Sciences – Istanbul, Türkiye

Taught courses in information technologies, management information systems, office programs, media literacy, and time management. Continued research in data science, machine learning, geographic information systems, and geospatial analysis while contributing to departmental administration, academic service, and curriculum development.

2021 – 2022

Research Assistant, PhD

Marmara University, Vocational School of Social Sciences – Istanbul, Türkiye

Supported teaching, research, publication, and departmental administrative activities following completion of the PhD.

Education

2014 – 2021

Istanbul Technical University – Istanbul, Türkiye

PhD in Applied Informatics

Geographical Information Technologies Program

Dissertation: *Spatio-Temporal Analysis of Applied Electronic Monitoring on Parolees of Mobility: A Case Study in Istanbul*

Supervisor: [Prof Dr Dursun Zafer Şeker](#)

2011 – 2013

Sakarya University – Sakarya, Türkiye

MSc in Information Systems

Management Information Systems Program

Supervisor: [Prof Dr Orhan Torkul](#)

2007 – 2011

Anadolu University – Eskişehir, Türkiye

BA in Business Administration

Peer-Reviewed Journal Articles

2026

Performance Comparison of Multimodal Vision-Language Models in Classifying Turkish Dishes [DOI](#)

Yunus Serhat Bıçakçı.

Sakarya University Journal of Computer and Information Sciences, 9(1), 119–133.

2025

Comparative Assessment of CNN and Transformer U-Nets in Multiple Sclerosis Lesion Segmentation [DOI](#)

Beytullah Sarıca, **Yunus Serhat Bıçakçı**, Dursun Zafer Şeker.

International Journal of Imaging Systems and Technology, 35(4), e70146.

2023

ATTransUNet: Semantic Segmentation Model for Building Segmentation from Aerial Image and Laser Data [DOI](#)

Yunus Serhat Bıçakçı, Beytullah Sarıca.

Nordic Machine Intelligence, 2, 7–9.

2020

Location-Based Analyses for Electronic Monitoring of Parolees [DOI](#)

Yunus Serhat Bıçakçı, Dursun Zafer Şeker, Hande Demirel.

ISPRS International Journal of Geo-Information, 9(5), 296.

2019

Spora Yönelik Tutum İle Yaşam Doyumu Arasındaki İlişki: Marmara Üniversitesi Sosyal Bilimler Meslek Yüksekokulu Öğrencileri Örneği [Link](#)

Çağdaş Caz, **Yunus Serhat Bıçakçı**, Fatma Firuzan Nakipoğlu.

Gümüşhane Üniversitesi Sosyal Bilimler Enstitüsü Elektronik Dergisi, 10(3), 484–491.

2017

Spatio-Temporal Analyses to Estimate Speed Information for Transport Forecasting and Scenario Testing [Link](#)

Yunus Serhat Bıçakçı, Beytullah Sarıca, Mete Ercan Pakdil, Batuhan Yazırlı, Hande Demirel.

Fresenius Environmental Bulletin, 26(1), 100–106.

Peer-Reviewed Conference Papers and Proceedings

- 2026 A Graph-Based Community Detection Approach for Identifying the Semantic Neighbourhoods within London's Airbnb Properties [DOI](#)
Joseph Shingleton, **Yunus Serhat Bıçakçı**, Yu Wang, Ana Basiri, Meiliu Wu.
AGILE: GIScience Series, 7, 41. Peer-reviewed short paper presented at the 29th AGILE Conference on Geographic Information Science, 16–19 June 2026, Tartu, Estonia. Best Short Paper Award.
- 2026 Mapping the Semantics of the Street: A VLM-Driven Geospatial Analysis in Fatih, Istanbul [DOI](#)
Yunus Serhat Bıçakçı, Joseph Shingleton, Yu Wang, Anahid Basiri.
Peer-reviewed full paper presented at the 1st International Conference on Geospatial Artificial Intelligence (GeoAI 2026), 3–6 June 2026, Ghent, Belgium.
- 2026 The Invisible City: Not All Streets Are Seen Equally [DOI](#)
Yu Wang, Joseph Shingleton, **Yunus Serhat Bıçakçı**, Anahid Basiri, James Ackland.
Peer-reviewed full paper presented at the 1st International Conference on Geospatial Artificial Intelligence (GeoAI 2026), 3–6 June 2026, Ghent, Belgium.
- 2026 Emergent Spatio-Semantic Structure in Large Language Model Embedding Spaces [Preprint DOI](#)
Joseph Shingleton, **Yunus Serhat Bıçakçı**, Yu Wang, Anahid Basiri.
Peer-reviewed paper presented orally at the 1st International Conference on Geospatial Artificial Intelligence (GeoAI 2026), 3–6 June 2026, Ghent, Belgium; associated EarthArXiv preprint available at the linked DOI.

Manuscripts Under Review and Revision

- 2026 When Does Detector Fusion Fail? A Systematic Benchmark of YOLO and DETR Ensembles for Military Vehicle Detection in UAV Imagery
Yunus Serhat Bıçakçı.
Signal, Image and Video Processing. Under editorial assessment (with editor).
- 2026 Impervious Surface Segmentation from Multisource Satellite Imagery Using Early-Fusion Deep Ensembles
Yunus Serhat Bıçakçı.
Journal manuscript under review; submitted 14 May 2026.
- 2026 Evaluating Explanation Integrity in a Vision Language Model for Urban Perception
Joseph Shingleton, **Yunus Serhat Bıçakçı**, Meiliu Wu, Yu Wang, Ana Basiri.
Transactions in GIS. Major revision requested; revised manuscript prepared for resubmission.
- 2026 A Use-Conditioned, Decision-Oriented Due-Diligence Protocol for Open AI-Derived Geospatial Products: Fields of The World over Türkiye
Yunus Serhat Bıçakçı.
Submitted to Earth Science Informatics on 31 July 2026.

Preprints

- 2026 Emergent Spatio-Semantic Structure in Large Language Model Embedding Spaces [DOI](#)

- Joseph Shingleton, **Yunus Serhat Bıçakçı**, Yu Wang, Anahid Basiri.
EarthArXiv preprint.
- 2026 Hybrid CNN-Transformer Ensemble for Enhanced Tank Detection in Aerial Imagery
[DOI](#)
Yunus Serhat Bıçakçı.
Research Square preprint.
- 2025 Street-Level Geolocalization Using Multimodal Large Language Models and Retrieval-Augmented Generation [DOI](#)
Yunus Serhat Bıçakçı, Joseph Shingleton, Anahid Basiri.
arXiv:2509.01341 [cs.CV].

Manuscripts in Preparation

- 2026 Encode Once, Ask Many Times: Street-Level Embeddings as Reusable Spatial Memory for GeoAI
Yunus Serhat Bıçakçı, Ana Basiri.
Manuscript in preparation; abstract submitted on 1 August 2026 for consideration in the International Journal of Geographical Information Science special issue "Critical Challenges in GeoAI".

Book Chapters

- 2026 Towards GeoAI Foundation Models: A Multimodal Learning Framework with Geospatial Intelligence
Meiliu Wu, Qunying Huang, Anahid Basiri, **Yunus Serhat Bıçakçı**, Joseph Shingleton.
In: Geography According to Foundation Models, Frontiers in Artificial Intelligence and Applications, Vol. 422, IOS Press, pp. 28–50. [DOI](#)
- 2025 Big Data and Social Media Analytics: Opportunities for Interactive Crime Mapping
Yunus Serhat Bıçakçı, Alina Ristea, Kate Bowers.
In: New Research in Crime Modeling and Mapping Using Geospatial Technologies, Leitner, M. (ed.), Geotechnologies and the Environment, Springer, pp. 3–25. [DOI](#)
- 2025 Yapay Zekanın Sokak Seviyesi Görüntü Analizinde Kullanımı
Yunus Serhat Bıçakçı.
In: Türkiye ve Dünyada Yönetim Bilişim Sistemleri, Tokmak, M. (ed.), YAZ Yayınları, Afyonkarahisar, pp. 77–93.
- 2022 Turizm ve Seyahat Bürolarında Yapay Zeka Uygulamaları
Yunus Serhat Bıçakçı, Bora Aksu.
In: Dijital Dönüşümün Turizme Yansımaları, Semiz Çelik, D. (ed.), Beta Yayınları, İstanbul.

Research Software and Open-Source Contributions

- 2026 GeoAI-VLM: Geospatial Vision-Language Model Analysis [DOI](#)
Yunus Serhat Bıçakçı.
Open-source research software for downloading Mapillary imagery by location and generating structured street-level descriptions using vision-language models. Zenodo archived release v0.3; [GitHub repository](#).
- 2026 sppt: Spatial Point Pattern Test for Aggregated Data (Python) [DOI](#)
Yunus Serhat Bıçakçı.

Open-source Python implementation of the Spatial Point Pattern Test for aggregated count data, using bootstrap resampling and S-Index metrics to compare spatial distributions. Based on the original R implementation by Martin A. Andresen; [GitHub repository](#).

Conference Papers and Presentations

- 2025 Digitalising Probation with Geospatial AI: Street-Level Human Perception in Istanbul's Fatih District
Yunus Serhat Bıçakçı, Alina Ristea.
Turkish Probation System 20th Year International Congress, 1–3 December 2025, Antalya, Türkiye. Summary text.
- 2025 Multimodal Semantic Segmentation for Impervious Surface Mapping
Yunus Serhat Bıçakçı.
15th Workshop on Hyperspectral Image and Signal Processing: Evolution of Remote Sensing (WHISPERS 2025), 12–14 November 2025, Bellaterra, Barcelona, Spain.
- 2025 From Video to Recipe: Leveraging Vision Language Models for Culinary Analysis
Yunus Serhat Bıçakçı.
4th International Congress on Food Researches, 16–18 October 2025, Sivas, Türkiye. Summary text.
- 2023 Using Big Data, Artificial Intelligence and Cloud Technologies for Interactive Crime Mapping [Link](#)
Yunus Serhat Bıçakçı, Alina Ristea, Kate Bowers.
Stockholm Criminology Symposium, 12–14 June 2023, Stockholm, Sweden.
- 2023 Application of Natural Language Processing Techniques for Crime Detection [Link](#)
Yunus Serhat Bıçakçı, Alina Ristea, Kate Bowers.
Environmental Criminology and Crime Analysis (ECCA) 2023, 14–16 June 2023, Stockholm, Sweden.
- 2022 Artificial Intelligence Applications for Social Municipality: The Case of Chatbot [Link](#)
Yunus Serhat Bıçakçı.
International VI. Civil Strategy Symposium, 8–9 December 2022, Istanbul, Türkiye.
- 2017 Büro Yönetimi ve Yönetici Asistanlığı Öğrencilerinin Beklentilerinin Belirlenmesi
Yunus Serhat Bıçakçı, Fatma Firuzan Nakipoğlu.
VI. International Vocational Schools Symposium, 18–20 May 2017, Sarajevo, Bosnia and Herzegovina. Full text.
- 2015 Elektronik İzleme Teknolojisi ve Türkiye’de Uygulanması
Yunus Serhat Bıçakçı, Dursun Zafer Şeker.
Türkiye’de Denetimli Serbestlik 10. Yıl Uluslararası Sempozyumu, 8–10 December 2015, Istanbul, Türkiye. Full text.

Research Funding and Fellowships

- 2026 – 2027 AI-Based Analysis of Istanbul Street-Level Imagery and Creation of an Open Dataset
Role: Principal Investigator; Program: Research Universities Support Program; Project ID: 12127.
Award: TRY 546,500; Duration: 12 months.

- Lead a multidisciplinary project to collect and analyse approximately three million crowdsourced and openly accessible street-level images covering Istanbul. The project applies GeoAI and computer vision methods, validates model-derived attributes, and will release a large-scale open research dataset supporting urban analytics, environmental studies, public safety, planning, and interdisciplinary research.
- 2026 SokakPuan – STEPS Open Innovation Call, Sokaklar Dönüşüyor Program
Role: Principal Investigator; *Award:* TRY 100,000.
Selected through the EIT Urban Mobility-supported STEPS (Smart Tools for Evaluating People-Centered Streets) initiative. Lead the development and municipal piloting of a human-centred street evaluation platform combining street-level imagery, GIS, AI-derived perception scores, and citizen feedback to assess accessibility, child safety, walkability, perceived safety, caregiver comfort, and public-space quality. The award supported the acquisition of two panoramic cameras for field deployment.
- 2026 – 2027 Investigating the Use of Panoramic Camera Imagery to Monitor Phenological Stages in Vineyards
Role: Researcher
Project period: August 2026–August 2027; *Duration:* 12 months.
Contribute to a collaborative precision-viticulture project using georeferenced 360-degree imagery, YOLO-based object detection, vision-language models, structure-from-motion photogrammetry, and 3D segmentation to monitor seasonal vine development at a research vineyard in Tekirdağ. Expected outputs include an annotated imagery dataset, phenological-stage detection models, 3D vine representations, and an interactive web-based visualisation workflow.
- 2023 – 2025 TÜBİTAK 2219 International Postdoctoral Research Fellowship
Role: Fellowship recipient and postdoctoral researcher.
Award: GBP 49,200; *Duration:* 2 years.
Competitive fellowship awarded by the Scientific and Technological Research Council of Türkiye to support postdoctoral research on AI-driven crime mapping, geospatial analysis, and urban data science at University College London.
- 2022 – 2024 Multi-Lingual and Multi-Modal Location Information Extraction (Multi-LM)
Role: Research Associate / Researcher; *Funder:* The Alan Turing Institute; *Grant No.:* 317199-2.
Award: GBP 394,911; *Project period:* May 2022–October 2024.
Contributed to application development and geolocation research at the University of Glasgow and co-authored a paper arising from the project's geolocation work.

Submitted Grant Applications

- 2026 Estimating Fundamental Housing Prices from Street-Level Imagery with Vision-Language Models: A Comparative Decision-Support System for Post-Earthquake and Ordinary Urban Contexts
Program: TÜBİTAK ARDEB 1005 – National New Ideas and Products Research Support Program; *Project No.:* 975877.
Role: Researcher; *Lead institution:* Hatay Mustafa Kemal University; *Status:* Submitted, *Award:* TRY 1,200,000, decision pending.
Proposed 12-month collaborative project to develop a housing-price estimation tool using Mapillary and 360-degree imagery, vision-language models, YOLO, SAM, and a mobile application across Istanbul, Izmir, and Hatay, including post-earthquake neighbourhoods in Antakya and Samandağ.

Awards and Honours

- 2026 **Best Short Paper Award – AGILE 2026**
Awarded for A Graph-Based Community Detection Approach for Identifying the Semantic Neighbourhoods within London's Airbnb Properties, 29th AGILE Conference on Geographic Information Science, Tartu, Estonia.
- 2025 **1st Place, WHISPERS 2025 ISASeg Challenge – Phase 1 and Phase 2**
Ranked first in both phases using multimodal deep-learning ensembles for impervious-surface segmentation from Sentinel-1 and Sentinel-2 imagery.
- 2024 – Present **Global Talent Endorsement – The British Academy**
Received endorsement under the UK's Global Talent visa scheme, recognising research promise and contribution in the relevant academic field.

Workshops

- 2026 State of the Art and Outlook of Street-Level Imagery – GeoAI 2026
In person, 3 June 2026, Ghent, Belgium.
- 2025 Spatial Data Science Symposium
Online.
- 2024 Spatial Data Science Symposium
Online.
- 2024 15th International Crime Science Conference
Online, 10 June 2024, London, UK.
- 2023 Mobile Phone Data for Mobility Workshop
Online, 6–7 December 2023, UK.
- 2023 Spatial Data Science Across Languages (SDSL)
In person, 18–19 September 2023, Münster, Germany.
- 2023 Scottish Crime Data Hackathon
Online, 25 July 2023, Edinburgh, Scotland.
- 2023 14th International Crime Science Conference
Online, 19 June 2023, London, UK.
- 2023 2nd Cardiff NLP Summer Workshop
In person, 26–27 June 2023, Cardiff, UK.

Teaching Experience

- 2026 – 2027 **Marmara University, Department of Artificial Intelligence and Machine Learning**
Scheduled for Fall 2026: Data Structures (YZM2005); Career Planning (YZM2053).
- 2025 – 2026 **Marmara University, Vocational School of Social Sciences**
Courses taught: Information and Communication Technologies (BSP1005; Fall and Spring); Office Applications (BSP1060, BSP1071, and BSP2007); Media Literacy (ILT2005); Management Information Systems (BYS2018 and ILT2006).
- 2024 – 2025 **Marmara University, Vocational School of Social Sciences**
Courses taught: Information and Communication Technologies (BSP1005); Management Information Systems (ILT2006); Telephone Communication and Time Management (BYS2034).
- 2022 – 2023 **Marmara University, Vocational School of Social Sciences**
Courses taught: Office Applications (BSP2007); Computer I (BSP205); Media Literacy (ILT2005).

Mentorship and Student Development

2023 – 2026

Mentor, inzva Mentorship Program

Mentored two third- or fourth-year undergraduate students in each annual cohort, providing individual guidance on academic development, research planning, technical skills, and careers in computer science and artificial intelligence.

2025–2026: Barkın Sağın and Kuntay Yılmaz

2024–2025: İsmail Talaz and Yunus Emre Erkan

2023–2024: Defne Ekin and Alper Solmaz

Doctoral Dissertation

2021

Spatio-Temporal Analysis of Applied Electronic Monitoring on Parolees of Mobility: A Case Study in Istanbul [Link](#)

Istanbul Technical University, Graduate School, Applied Informatics, Geographical Information Technologies Program.

University and Professional Service

2026 – Present

GIScience 2027 – Fourteenth International Conference on Geographic Information Science

International Committee Chair.

2023 – Present

Environmental Criminology and Crime Analysis (ECCA)

Delegate.

2026 - 2026

Journal Referee

International Journal of Environment and Geoinformatics (IJEGEO).

2023 – 2025

Marmara University, Vocational School of Social Sciences

Vice Head of Department, Department of Office Technologies and Data Management.

2022 - 2022

Editorial Board / Co-Editor

The Social Research Journal of Marmara University.

2021 – 2022

Journal Referee

International Humanities and Social Science Review (IHSSR).

Talks and Certifications

1st Place Award, Multimodal Segmentation Challenge [Certificate](#)

Hugging Face Agents Course; 06/2025 [Certificate](#)

Deep Learning Specialization, Coursera / DeepLearning.AI [Credential](#)

TensorFlow Developer, DeepLearning.AI [Credential](#)

NVIDIA: Fundamentals of Deep Learning [Certificate](#)

Intro to Machine Learning, Talk – Google DSC Işık University; 11/2022 [Video](#)

TensorFlow Basics, Talk – Google DSC Bandırma University; 12/2022 [Video](#)

Google Cloud Skills Boost [Profile](#); Google Developer [Page](#)

DataCamp: Intermediate R [Certificate](#)

Skills

Programming – Python, R, SQL, Bash

Data and Geospatial Tools – OpenCV, NumPy, Pandas, GeoPandas, DuckDB, PostGIS, QGIS, Esri ArcGIS

Generative AI and Inference – Transformers, vLLM, LMDeploy, Ollama, OpenAI API, quantisation, AWQ, GPTQ, GGUF

RAG and Agentic Systems – Retrieval-Augmented Generation, MCP, LlamaIndex, LangGraph, FAISS, vector databases

Computer Vision – Vision Transformers, CLIP, SigLIP, DINO, semantic segmentation, object detection, image classification

Multimodal AI – Vision-language models, multimodal evaluation, dataset curation, prompt-based analysis, visual reasoning

Remote Sensing and GeoAI – Sentinel-1, Sentinel-2, SAR-optical fusion, street-level imagery, geospatial foundation models, spatial analytics

MLOps and Deployment – Docker, Git, CI/CD, model serving, cloud inference, experiment tracking, monitoring

Cloud and Platforms – Google Cloud Platform, Microsoft Azure, Hugging Face, Kaggle, GitHub